Lecture 7

Cartan subalgebras and root decomposition

Objectives. *We use simultaneous eigenspaces to study the structure of a semisimple Lie algebra. We take a maximal subalgebra of elements x with $\operatorname{ad} x$ diagonalisable. Its adjoint eigenspaces for nonzero weights are the root spaces. The Killing form pairs opposite root spaces.*

By the end of this lecture you will be able to:

- Prove that toral subalgebras are abelian and that Cartan subalgebras exist.
- Derive the root decomposition and the rule $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$.
- Prove the Killing-form orthogonality relations.
- Compute the roots, root matrices and brackets of $\mathfrak{sl}_3(\mathbb{C})$.

Throughout, $\mathfrak{g}$ is a nonzero semisimple complex Lie algebra with Killing form κ. By Lecture 4, κ is symmetric, nondegenerate and invariant:

$$\kappa([x,y],z) = \kappa(x,[y,z]) \quad \text{for all } x,y,z \in \mathfrak{g}.$$

The centre $Z(\mathfrak{g})$ is an abelian ideal. Abelian ideals are solvable, so $Z(\mathfrak{g})$ lies in the radical, which is zero. Hence

$$Z(\mathfrak{g}) = 0.$$

7.1 Toral subalgebras and imports

An element $x \in \mathfrak{g}$ is *semisimple* if $\operatorname{ad} x$ is diagonalisable. The element x is *nilpotent* if $\operatorname{ad} x$ is nilpotent. Lecture 4 stated the internal Jordan decomposition [C] (Humphreys §5.4): each $x \in \mathfrak{g}$ has a unique expression

$$x = x_s + x_n, \qquad x_s, x_n \in \mathfrak{g}, \qquad [x_s, x_n] = 0,$$

with x_s semisimple and x_n nilpotent. If x is semisimple, then $x = x + 0$ is such an expression. Uniqueness gives $x_s = x$ and $x_n = 0$.

Definition 7.1 (Toral and Cartan subalgebras)**.** A subalgebra $\mathfrak{t} \subseteq \mathfrak{g}$ is *toral* if every element of $\mathfrak{t}$ is semisimple. A *Cartan subalgebra* of $\mathfrak{g}$ is a toral subalgebra not properly contained in another toral subalgebra.

The zero subspace is toral. Every toral subalgebra has dimension at most $\dim \mathfrak{g}$, so it lies in a maximal one. Hence Cartan subalgebras exist.

The next proof uses a fact from linear algebra. Let T be a diagonalisable operator with distinct eigenvalues $c_1, \dots, c_r$, and let U be a T-invariant subspace. Then

$$(T - c_1) \cdots (T - c_r) = 0 \quad \Longrightarrow \quad (T|_U - c_1) \cdots (T|_U - c_r) = 0.$$

So the minimal polynomial of $T|_U$ divides a product of distinct linear factors. Hence $T|_U$ is diagonalisable.

Lemma 7.2 (Toral subalgebras are abelian · *[P]*)**.** *Every toral subalgebra $\mathfrak{t}$ of $\mathfrak{g}$ is abelian.*

Proof. Fix $x \in \mathfrak{t}$. Let $\operatorname{ad}_{\mathfrak{t}} x$ be the restriction of $\operatorname{ad} x$ to the invariant subspace $\mathfrak{t}$. By the fact above, this restriction is diagonalisable. A diagonalisable operator whose only eigenvalue is 0 is zero. So it suffices to show that $\operatorname{ad}_{\mathfrak{t}} x$ has no nonzero eigenvalue.

Suppose that $y \in \mathfrak{t}$ is nonzero and $[x, y] = ay$ with $a \neq 0$. Apply $\operatorname{ad}_{\mathfrak{t}} y$ twice:

$$(\operatorname{ad}_{\mathfrak{t}} y)(x) = [y, x] = -ay,$$
$$(\operatorname{ad}_{\mathfrak{t}} y)(-ay) = -a[y, y] = 0.$$

Thus the vector $[y, x]$ lies in the kernel of $\operatorname{ad}_{\mathfrak{t}} y$.

The operator $\operatorname{ad}_{\mathfrak{t}} y$ is diagonalisable as well. Decompose x into eigenvectors of $\operatorname{ad}_{\mathfrak{t}} y$:

$$x = \sum_{\lambda} x_\lambda, \qquad [y, x_\lambda] = \lambda x_\lambda.$$

Applying $\operatorname{ad}_{\mathfrak{t}} y$ gives

$$[y, x] = \sum_{\lambda \neq 0} \lambda x_\lambda.$$

The right side lies in the sum of the nonzero eigenspaces. That sum meets the kernel only in 0. Hence

$$-ay = [y, x] = 0.$$

Since $a \neq 0$ and $y \neq 0$, this is a contradiction. ∎

We import two theorems here, and a third in Section 7.5. The second is used only in Corollary 7.9.

Theorem 7.3 (Self-centraliser theorem · *[C]*)**.** *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and let $\mathfrak{h}$ be a Cartan subalgebra. Then*

$$C_{\mathfrak{g}}(\mathfrak{h}) = \{x \in \mathfrak{g} : [h, x] = 0 \text{ for all } h \in \mathfrak{h}\} = \mathfrak{h}.$$

Source: Humphreys §8.2, Proposition, for a semisimple algebra over an algebraically closed field of characteristic zero. The inclusion $\mathfrak{h} \subseteq C_{\mathfrak{g}}(\mathfrak{h})$ is Lemma 7.2. The reverse inclusion is the imported part.

Hence every Cartan subalgebra is nonzero, because the zero subalgebra has centraliser $C_{\mathfrak{g}}(0) = \mathfrak{g} \neq 0$.

Theorem 7.4 (Jordan decomposition in representations · *[C]*)**.** *Let $\mathfrak{g}$ be semisimple, let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation, and let $x = x_s + x_n$ be the internal Jordan decomposition of $x \in \mathfrak{g}$. Then $\rho(x_s)$ is diagonalisable, $\rho(x_n)$ is nilpotent, and $\rho(x) = \rho(x_s) + \rho(x_n)$ is the Jordan decomposition of the operator $\rho(x)$.*

Source: Humphreys §6.4, Corollary. Its proof uses Weyl's theorem from Lecture 5.

7.2 The root decomposition

Before the general argument, here is a short preview. It is taken from the worked example in §7.4, and no proof below uses it.

In $\mathfrak{sl}_2$ the operator $\operatorname{ad} h$ is diagonal in the basis (h, e, f). By Lecture 6, the weights of h, e and f are 0, 2 and -2. In $\mathfrak{sl}_3$ the traceless diagonal matrices take the place of the single element h. For $H = \operatorname{diag}(a_1, a_2, a_3)$ with $a_1 + a_2 + a_3 = 0$ and for $i \neq j$,

$$[H, E_{ij}] = (a_i - a_j)E_{ij}.$$

So each matrix unit E_{ij} with $i \neq j$ is an eigenvector of every $\operatorname{ad} H$ at once. Its eigenvalue $a_i - a_j$ depends linearly on H, so it is a linear form on the space of these diagonal matrices. Diagonal matrices commute.

We seek the corresponding diagonalising subalgebra in a general semisimple algebra. A Cartan subalgebra plays this role, and the eigenvalues become linear forms on it.

Lemma 7.5 (Simultaneous eigenspaces · *[P]*). *Let $\mathfrak{a}$ be an abelian Lie algebra, and let ρ be a representation of $\mathfrak{a}$ on a finite-dimensional space V such that every $\rho(a)$ is diagonalisable. For $\mu \in \mathfrak{a}^*$ put*

$$V_\mu = \{v \in V : \rho(a)v = \mu(a)v \text{ for all } a \in \mathfrak{a}\}.$$

Then $V = \bigoplus_{\mu \in \mathfrak{a}^*} V_\mu$, and only finitely many summands are nonzero.

Proof. The operators commute, because $\mathfrak{a}$ is abelian:

$$\rho(a)\rho(b) - \rho(b)\rho(a) = \rho([a, b]) = 0.$$

Choose a basis $T_1, \dots, T_m$ of the finite-dimensional space $\rho(\mathfrak{a})$. These commuting diagonalisable operators have a common eigenbasis $(v_1, \dots, v_d)$. It is an eigenbasis for every $\rho(a)$, since each $\rho(a)$ is a linear combination of the T_i. Write $\rho(a)v_k = \mu_k(a)v_k$. For $a, b \in \mathfrak{a}$ and $s, t \in \mathbb{C}$,

$$\begin{aligned} \mu_k(sa + tb)\, v_k &= \rho(sa + tb)v_k \\ &= s\rho(a)v_k + t\rho(b)v_k = \big(s\mu_k(a) + t\mu_k(b)\big)v_k. \end{aligned}$$

Thus each μ_k is linear, $v_k \in V_{\mu_k}$, and the spaces V_μ span V.

For directness, let $v = \sum_k c_k v_k \in V_\mu$. For every $a \in \mathfrak{a}$,

$$0 = \big(\rho(a) - \mu(a)\big)v = \sum_{k=1}^{d} c_k \big(\mu_k(a) - \mu(a)\big)v_k.$$

So $c_k = 0$ whenever $\mu_k \neq \mu$. Thus V_μ is spanned by the v_k with $\mu_k = \mu$. The sum is direct, and only the forms μ_k occur. ∎

Fix a Cartan subalgebra $\mathfrak{h}$. It is abelian by Lemma 7.2, and every $\operatorname{ad} h$ with $h \in \mathfrak{h}$ is diagonalisable. Apply Lemma 7.5 to the representation $\mathfrak{h} \to \mathfrak{gl}(\mathfrak{g})$, $h \mapsto \operatorname{ad} h$.

Definition 7.6 (Roots). For $\alpha \in \mathfrak{h}^*$, put

$$\mathfrak{g}_\alpha = \{x \in \mathfrak{g} : [h, x] = \alpha(h)x \text{ for all } h \in \mathfrak{h}\}.$$

A *root* is a nonzero $\alpha \in \mathfrak{h}^*$ with $\mathfrak{g}_\alpha \neq 0$. For a root α, the space $\mathfrak{g}_\alpha$ is its *root space*. We write R for the set of roots.

Theorem 7.7 (Root decomposition · *[P]*). *The set R is finite, and*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha .$$

Proof. Lemma 7.5 gives

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha ,$$

with R finite. By definition, $\mathfrak{g}_0$ is the centraliser of $\mathfrak{h}$. Theorem 7.3 identifies it:

$$\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

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Proposition 7.8 (Bracket rule · *[P]*). *For all $\alpha, \beta \in \mathfrak{h}^*$,*

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

Proof. Let $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$. The Jacobi identity in derivation form (Lecture 1) gives

$$\begin{aligned} [h,[x,y]] &= [[h,x],y] + [x,[h,y]] \\ &= \alpha(h)[x,y] + \beta(h)[x,y] = (\alpha+\beta)(h)\,[x,y]. \end{aligned}$$

■

Three consequences follow.

(i) If $\alpha + \beta \notin R \cup \{0\}$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = 0$.

(ii) If $\alpha \in R$, then $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq \mathfrak{g}_0 = \mathfrak{h}$.

(iii) If $\alpha \in R$ and $x \in \mathfrak{g}_\alpha$, then $\operatorname{ad} x$ is nilpotent.

For (iii), apply the bracket rule k times:

$$(\operatorname{ad} x)^k \mathfrak{g}_\beta \subseteq \mathfrak{g}_{\beta+k\alpha} \quad \text{for all } k \geq 0.$$

Since $\alpha \neq 0$, the forms $\beta + k\alpha$ with $k \geq 0$ are distinct. Only finitely many lie in the finite set $R \cup \{0\}$. So for each $\beta \in R \cup \{0\}$ there is N_β with $(\operatorname{ad} x)^k \mathfrak{g}_\beta = 0$ for $k \geq N_\beta$. Let N be the largest N_β. By the root decomposition,

$$(\operatorname{ad} x)^N \mathfrak{g} = \sum_{\beta \in R \cup \{0\}} (\operatorname{ad} x)^N \mathfrak{g}_\beta = 0.$$

Corollary 7.9 (Weights of a module · *[P] given Theorem 7.4*). *Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a finite-dimensional representation. Define V_μ as in Lemma 7.5 for ρ restricted to $\mathfrak{h}$. Then*

$$V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu, \qquad \rho(\mathfrak{g}_\alpha) V_\mu \subseteq V_{\mu+\alpha} \quad \text{for all } \alpha, \mu \in \mathfrak{h}^*.$$

Proof. For $h \in \mathfrak{h}$ we have $h_s = h$, so $\rho(h)$ is diagonalisable by Theorem 7.4. Lemma 7.5 gives the decomposition.

For $x \in \mathfrak{g}_\alpha$ and $v \in V_\mu$, use $\rho([h,x]) = \alpha(h)\rho(x)$:

$$\begin{aligned} \rho(h)\rho(x)v &= \rho(x)\rho(h)v + \rho([h,x])v \\ &= \mu(h)\rho(x)v + \alpha(h)\rho(x)v = \big(\mu+\alpha\big)(h)\,\rho(x)v. \end{aligned}$$

■

The forms μ with $V_\mu \neq 0$ are the *weights* of V. For the adjoint representation they are 0 and the roots.

7.3 Killing-form orthogonality

Proposition 7.10 (Orthogonality · *[P]*)**.**

(a) If $\alpha, \beta \in \mathfrak{h}^*$ and $\alpha + \beta \neq 0$, then $\kappa(\mathfrak{g}_\alpha, \mathfrak{g}_\beta) = 0$.

(b) The restriction of κ to $\mathfrak{h}$ is nondegenerate.

(c) If $\alpha \in R$, then $-\alpha \in R$, and κ restricts to a nondegenerate pairing $\mathfrak{g}_\alpha \times \mathfrak{g}_{-\alpha} \to \mathbb{C}$.

Proof. (a) Let $x \in \mathfrak{g}_\alpha$, $y \in \mathfrak{g}_\beta$ and $h \in \mathfrak{h}$. By antisymmetry and invariance,

$$\begin{aligned} \alpha(h)\,\kappa(x,y) &= \kappa([h,x],y) = -\kappa([x,h],y) \\ &= -\kappa(x,[h,y]) = -\beta(h)\,\kappa(x,y). \end{aligned}$$

Thus $(\alpha+\beta)(h)\,\kappa(x,y) = 0$ for every $h \in \mathfrak{h}$. Since $\alpha+\beta \neq 0$, some h has $(\alpha+\beta)(h) \neq 0$. Hence $\kappa(x,y) = 0$.

(b) Let $z \in \mathfrak{h}$ satisfy $\kappa(z,\mathfrak{h}) = 0$. Since $z \in \mathfrak{g}_0$, part (a) gives $\kappa(z,\mathfrak{g}_\beta) = 0$ for every root β. Take $y \in \mathfrak{g}$ and decompose it by Theorem 7.7:

$$y = y_0 + \sum_{\beta \in R} y_\beta, \qquad y_0 \in \mathfrak{h}, \quad y_\beta \in \mathfrak{g}_\beta.$$

Then

$$\kappa(z,y) = \kappa(z,y_0) + \sum_{\beta \in R} \kappa(z,y_\beta) = 0.$$

Nondegeneracy of κ on $\mathfrak{g}$ gives $z = 0$.

(c) Let $x \in \mathfrak{g}_\alpha$ be nonzero. Nondegeneracy of κ gives $y \in \mathfrak{g}$ with $\kappa(x,y) \neq 0$. Decompose y as in (b). By (a), $\kappa(x,y_0) = 0$ because $\alpha \neq 0$, and $\kappa(x,y_\beta) = 0$ unless $\beta = -\alpha$. If $-\alpha \notin R$, every term vanishes, which is impossible. So $-\alpha \in R$, and

$$0 \neq \kappa(x,y) = \kappa(x,y_0) + \sum_{\beta \in R} \kappa(x,y_\beta) = \kappa(x,y_{-\alpha}).$$

Hence no nonzero element of $\mathfrak{g}_\alpha$ is orthogonal to $\mathfrak{g}_{-\alpha}$. Exchanging α and $-\alpha$ proves nondegeneracy on the other side. ∎

A nondegenerate pairing identifies each space with the dual of the other. Hence $\dim \mathfrak{g}_\alpha = \dim \mathfrak{g}_{-\alpha}$.

Since κ is nondegenerate on $\mathfrak{h}$, the linear map

$$\mathfrak{h} \longrightarrow \mathfrak{h}^*, \qquad t \longmapsto \kappa(t,\cdot)$$

has zero kernel. Both spaces have dimension $\dim \mathfrak{h}$, so the map is bijective. For $\lambda \in \mathfrak{h}^*$, let $t_\lambda \in \mathfrak{h}$ be the unique element with

$$\kappa(t_\lambda, h) = \lambda(h) \quad \text{for all } h \in \mathfrak{h}.$$

Proposition 7.11 (Roots and the Killing form · *[P]*)**.**

(a) The roots span $\mathfrak{h}^*$.

(b) If $\alpha \in R$, $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$, then $[x,y] = \kappa(x,y)\,t_\alpha$. Hence $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}t_\alpha$.

Proof. (a) Suppose the roots span a proper subspace of $\mathfrak{h}^*$. By duality, some nonzero $h \in \mathfrak{h}$ satisfies $\alpha(h) = 0$ for all $\alpha \in R$. Then

$$[h,x] = \alpha(h)x = 0 \qquad \text{for all } x \in \mathfrak{g}_\alpha,\ \alpha \in R.$$

Also $[h, \mathfrak{h}] = 0$, since $\mathfrak{h}$ is abelian. By Theorem 7.7, these subspaces span $\mathfrak{g}$. So $h \in Z(\mathfrak{g}) = 0$, a contradiction.

(b) By the bracket rule, $[x, y] \in \mathfrak{g}_0 = \mathfrak{h}$. For $h \in \mathfrak{h}$, invariance and the definition of t_α give

$$\begin{aligned}\kappa(h, [x, y]) &= \kappa([h, x], y) = \alpha(h)\,\kappa(x, y) \\ &= \kappa(t_\alpha, h)\,\kappa(x, y) = \kappa\big(h, \kappa(x, y)\, t_\alpha\big).\end{aligned}$$

Both $[x, y]$ and $\kappa(x, y)t_\alpha$ lie in $\mathfrak{h}$. Their difference is orthogonal to $\mathfrak{h}$. Since κ is nondegenerate on $\mathfrak{h}$ by Proposition 7.10(b), the difference is zero.

By Proposition 7.10(c), some $x \in \mathfrak{g}_\alpha$ and $y \in \mathfrak{g}_{-\alpha}$ have $\kappa(x, y) \neq 0$. Since $\alpha \neq 0$, also $t_\alpha \neq 0$. These brackets therefore span $\mathbb{C}t_\alpha$. ∎

Lecture 8 starts from Proposition 7.11(b). It attaches an $\mathfrak{sl}_2$-subalgebra to each root and proves $\dim \mathfrak{g}_\alpha = 1$.

7.4 Worked example: $\mathfrak{sl}_3(\mathbb{C})$

Let $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, and let $\mathfrak{h}$ be the traceless diagonal matrices, with basis

$$h_1 = E_{11} - E_{22}, \qquad h_2 = E_{22} - E_{33}.$$

Take $H = \operatorname{diag}(a_1, a_2, a_3)$ in $\mathfrak{h}$ and $X \in \mathfrak{g}$. The entries of the two products are

$$(HX)_{ij} = a_i X_{ij}, \qquad (XH)_{ij} = X_{ij} a_j.$$

Subtracting gives

$$[H, X]_{ij} = (a_i - a_j)X_{ij}, \qquad [H, E_{ij}] = (a_i - a_j)E_{ij}.$$

Also $[H, h_1] = [H, h_2] = 0$. The eight matrices h_1, h_2 and E_{ij} with $i \neq j$ form an eigenbasis of $\operatorname{ad} H$. Hence $\mathfrak{h}$ is toral.

Next, $\mathfrak{h}$ is a Cartan subalgebra. Put $H_0 = \operatorname{diag}(1, 0, -1)$. Its entry differences are

$$a_1 - a_2 = 1, \qquad a_1 - a_3 = 2, \qquad a_2 - a_3 = 1.$$

None is zero. If $X \in C_{\mathfrak{g}}(\mathfrak{h})$, then $[H_0, X] = 0$, so $X_{ij} = 0$ for $i \neq j$. Thus X is diagonal and traceless, and

$$C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}.$$

If $\mathfrak{t} \supseteq \mathfrak{h}$ is toral, Lemma 7.2 gives $\mathfrak{t} \subseteq C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. So $\mathfrak{t} = \mathfrak{h}$.

The proofs in §7.2 and §7.3 use Theorem 7.3 only through $\mathfrak{g}_0 = C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$. The required equality has been checked directly here.

Define $\varepsilon_i \in \mathfrak{h}^*$ by $\varepsilon_i(H) = a_i$. On $\mathfrak{h}$, $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$. The eigenvector E_{ij} has weight $\varepsilon_i - \varepsilon_j$. Put $\alpha_1 = \varepsilon_1 - \varepsilon_2$ and $\alpha_2 = \varepsilon_2 - \varepsilon_3$, so that $\varepsilon_1 - \varepsilon_3 = \alpha_1 + \alpha_2$. Two representative values are

$$\begin{aligned}\alpha_1(h_1) &= \varepsilon_1(h_1) - \varepsilon_2(h_1) = 1 - (-1) = 2, \\ \alpha_2(h_1) &= \varepsilon_2(h_1) - \varepsilon_3(h_1) = -1 - 0 = -1.\end{aligned}$$

The other values are found in the same way. The table lists the six weights, their root matrices and their values.

root	α_1	α_2	$\alpha_1+\alpha_2$	$-\alpha_1$	$-\alpha_2$	$-\alpha_1-\alpha_2$
root matrix	E_{12}	E_{23}	E_{13}	E_{21}	E_{32}	E_{31}
value at h_1	2	-1	1	-2	1	-1
value at h_2	-1	2	1	1	-2	-1

The table gives distinct nonzero value pairs. So the eigenbasis gives

$$R = \{\varepsilon_i - \varepsilon_j : i \neq j\}, \qquad \mathfrak{g}_{\varepsilon_i-\varepsilon_j} = \mathbb{C}E_{ij}, \qquad \dim \mathfrak{g} = 2 + 6.$$

Here one-dimensionality is read off from a basis. Lecture 8 proves it for every semisimple $\mathfrak{g}$.

Figure 7.1 draws the table. On its left, each root matrix stands at its position in a 3×3 matrix. The diagonal cells together stand for the single two-dimensional space $\mathfrak{h}$.

On its right, each root is a point of a plane. A root is a linear form on $\mathfrak{h}$, so it is determined by its values (a, b) at h_1 and h_2, which are the last two rows of the table. We plot the pair (a, b) at the point $a\,p + b\,q$, where p and q are two fixed plane vectors of equal length at an angle of 60 degrees. These coordinates are chosen for display. Adding linear forms adds their value pairs, so sums of roots are correct in the picture. Lengths and angles in the picture have no meaning yet. Lecture 8 constructs the invariant inner product on the real span of the roots, and Lecture 9 draws the roots with their true lengths and angles.

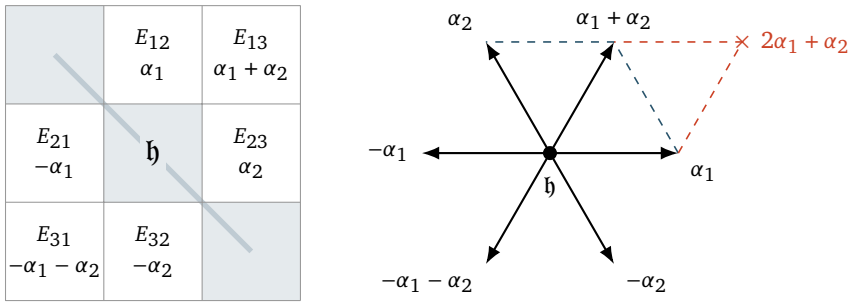


Figure 7.1: The roots of $\mathfrak{sl}_3(\mathbb{C})$ from matrix positions, plotted in coordinates chosen for display. The blue dashed lines add α_1 and α_2, the weights of E_{12} and E_{23}, and $[E_{12}, E_{23}] = E_{13}$. The red dashed lines add α_1 and $\alpha_1 + \alpha_2$. Their sum, marked $\times$, is neither a root nor zero, so $[E_{12}, E_{13}] = 0$.

A bracket of root matrices adds weights. If the sum of the two weights is zero, the bracket lies in $\mathfrak{h}$. If the sum is neither a root nor zero, the bracket vanishes. We now check these statements on examples.

The matrix-unit rule $E_{ij}E_{kl} = \delta_{jk}E_{il}$ gives four nonzero brackets:

$$\begin{aligned}
[E_{12}, E_{23}] &= E_{12}E_{23} - E_{23}E_{12} = E_{13} - 0 = E_{13},\\
[E_{13}, E_{32}] &= E_{13}E_{32} - E_{32}E_{13} = E_{12} - 0 = E_{12},\\
[E_{23}, E_{31}] &= E_{23}E_{31} - E_{31}E_{23} = E_{21} - 0 = E_{21},\\
[E_{12}, E_{21}] &= E_{12}E_{21} - E_{21}E_{12} = E_{11} - E_{22} = h_1.
\end{aligned}$$

Proposition 7.8 predicts that each result has the sum of the two input weights. In order, these sums are

$$\begin{aligned}
&\alpha_1 + \alpha_2, && (\alpha_1 + \alpha_2) - \alpha_2 = \alpha_1,\\
&\alpha_2 - (\alpha_1 + \alpha_2) = -\alpha_1, && \alpha_1 - \alpha_1 = 0.
\end{aligned}$$

They are the weights of E_{13}, E_{12}, E_{21} and h_1.

By contrast,

$$[E_{12}, E_{13}] = E_{12}E_{13} - E_{13}E_{12} = 0 - 0 = 0.$$

The bracket rule requires this, since the bracket lies in $\mathfrak{g}_{2\alpha_1+\alpha_2}$. The sum of the two weights takes the values

$$(2\alpha_1 + \alpha_2)(h_1) = 2 \cdot 2 - 1 = 3, \qquad (2\alpha_1 + \alpha_2)(h_2) = 2 \cdot (-1) + 2 = 0.$$

These values are not $(0, 0)$, and no column of the table has the values $(3, 0)$. So $2\alpha_1 + \alpha_2 \notin R \cup \{0\}$, and $\mathfrak{g}_{2\alpha_1+\alpha_2} = 0$. The cross in Figure 7.1 marks this sum.

Now compute κ on $\mathfrak{h}$. Take $H = \operatorname{diag}(a_i)$ and $H' = \operatorname{diag}(b_i)$ in $\mathfrak{h}$. The operator $\operatorname{ad} H \operatorname{ad} H'$ is zero on $\mathfrak{h}$ and multiplies E_{ij} by $(a_i - a_j)(b_i - b_j)$. Its trace in the eigenbasis is

$$\kappa(H, H') = \sum_{i \neq j} (a_i - a_j)(b_i - b_j) = \sum_{i,j=1}^{3} (a_i - a_j)(b_i - b_j),$$

since the terms with $i = j$ are zero. Expand the product and sum over i and j:

$$\begin{aligned} \sum_{i,j=1}^{3} (a_i - a_j)(b_i - b_j) &= \sum_{i,j=1}^{3} (a_i b_i + a_j b_j) - \sum_{i,j=1}^{3} (a_i b_j + a_j b_i) \\ &= 6 \sum_{i=1}^{3} a_i b_i - 2\Big(\sum_{i=1}^{3} a_i\Big)\Big(\sum_{j=1}^{3} b_j\Big) \\ &= 6 \sum_{i=1}^{3} a_i b_i. \end{aligned}$$

The last step uses $\sum_i a_i = \sum_i b_i = 0$.

Let $\alpha = \varepsilon_i - \varepsilon_j$ with $i \neq j$. The matrix $\frac{1}{6}(E_{ii} - E_{jj})$ lies in $\mathfrak{h}$, and

$$\kappa\Big(\frac{1}{6}(E_{ii} - E_{jj}), H\Big) = 6\Big(\frac{1}{6}a_i - \frac{1}{6}a_j\Big) = a_i - a_j = \alpha(H).$$

By uniqueness of t_α,

$$t_\alpha = \frac{1}{6}(E_{ii} - E_{jj}), \qquad t_{\alpha_1} = \frac{1}{6}h_1.$$

Proposition 7.11(b) now gives a Killing value without a trace:

$$h_1 = [E_{12}, E_{21}] = \kappa(E_{12}, E_{21})\, t_{\alpha_1} = \frac{1}{6}\kappa(E_{12}, E_{21})\, h_1.$$

Since $h_1 \neq 0$, we get $\kappa(E_{12}, E_{21}) = 6$. Proposition 7.10(a) gives other values with no calculation:

$$\kappa(E_{12}, E_{12}) = 0 \quad \text{since } 2\alpha_1 \neq 0,$$
$$\kappa(E_{12}, E_{23}) = 0 \quad \text{since } \alpha_1 + \alpha_2 \neq 0.$$

7.5 Changing the Cartan subalgebra

The root decomposition depends on the choice of $\mathfrak{h}$. An automorphism φ of $\mathfrak{g}$ carries the decomposition for $\mathfrak{h}$ to the decomposition for $\varphi(\mathfrak{h})$. For $x, y \in \mathfrak{g}$,

$$(\operatorname{ad} \varphi(x))(\varphi(y)) = [\varphi(x), \varphi(y)] = \varphi([x, y]), \quad \text{so} \quad \operatorname{ad} \varphi(x) = \varphi \circ \operatorname{ad} x \circ \varphi^{-1}.$$

Hence $\operatorname{ad}\varphi(x)$ is diagonalisable exactly when $\operatorname{ad} x$ is. The product $\operatorname{ad}\varphi(x)\operatorname{ad}\varphi(y)$ is conjugate to $\operatorname{ad} x \operatorname{ad} y$, so taking traces gives $\kappa(\varphi x, \varphi y) = \kappa(x, y)$.

Thus φ and φ^{-1} map toral subalgebras to toral subalgebras. If $\mathfrak{t}' \supseteq \varphi(\mathfrak{h})$ is toral, then $\varphi^{-1}(\mathfrak{t}') \supseteq \mathfrak{h}$ is toral. By maximality, $\varphi^{-1}(\mathfrak{t}') = \mathfrak{h}$, so $\mathfrak{t}' = \varphi(\mathfrak{h})$. Hence $\mathfrak{h}' = \varphi(\mathfrak{h})$ is a Cartan subalgebra.

For $x \in \mathfrak{g}_\alpha$ and $h \in \mathfrak{h}$,

$$[\varphi(h), \varphi(x)] = \varphi([h, x]) = \alpha(h)\,\varphi(x).$$

Put $\alpha' = \alpha \circ (\varphi|_{\mathfrak{h}})^{-1} \in (\mathfrak{h}')^*$. Hence $\varphi(\mathfrak{g}_\alpha)$ lies in the root space of $\mathfrak{h}'$ for α'. The same argument for φ^{-1} gives equality. Roots, brackets and Killing values are carried over unchanged.

Theorem 7.12 (Conjugacy of Cartan subalgebras · *[C]*)**.** *Let $\mathfrak{g}$ be a semisimple complex Lie algebra, and let $\mathfrak{h}$ and $\mathfrak{h}'$ be Cartan subalgebras. Then some automorphism φ of $\mathfrak{g}$ satisfies $\varphi(\mathfrak{h}) = \mathfrak{h}'$.*

Source: Humphreys §16.4, Corollary (conjugacy under the group $E(L)$). Humphreys defines Cartan subalgebras as nilpotent self-normalising subalgebras. By Humphreys §15.3, Corollary, these are exactly the maximal toral subalgebras when $\mathfrak{g}$ is semisimple.

The theorem and the transport argument show that the root decompositions of any two Cartan subalgebras are related by an automorphism of $\mathfrak{g}$. Their common dimension is the *rank* of $\mathfrak{g}$. Thus the classification in Lecture 10 does not depend on the choice.

Self-checks

(1) Use the basis (h, e, f) of $\mathfrak{sl}_2(\mathbb{C})$ from Lecture 1. Is the line $\mathbb{C}e$ toral?

(2) In $\mathfrak{sl}_3(\mathbb{C})$, which root space pairs nondegenerately with $\mathfrak{g}_{\alpha_1+\alpha_2}$? What is $[E_{13}, E_{31}]$?

(3) Can $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta]$ be nonzero when $\alpha + \beta \notin R \cup \{0\}$?

Answers. (1) No. Since $(\operatorname{ad} e)^2 f = [e, h] = -2e \neq 0$ and $(\operatorname{ad} e)^3 = 0$, the operator $\operatorname{ad} e$ is nonzero and nilpotent, hence not diagonalisable.
(2) The space $\mathfrak{g}_{-\alpha_1-\alpha_2} = \mathbb{C}E_{31}$. The bracket is

$$[E_{13}, E_{31}] = E_{11} - E_{33} = h_1 + h_2 = 6\,t_{\alpha_1+\alpha_2},$$

so $\kappa(E_{13}, E_{31}) = 6$.
(3) No. Such a bracket lies in $\mathfrak{g}_{\alpha+\beta} = 0$. For example, $[E_{12}, E_{13}] = 0$.

Summary

A Cartan subalgebra is a maximal toral subalgebra. It is abelian, and by the cited self-centraliser theorem it equals its centraliser. Its adjoint eigenspaces give

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha, \qquad [\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}.$$

The Killing form pairs $\mathfrak{g}_\alpha$ only with $\mathfrak{g}_{-\alpha}$, and it is nondegenerate on $\mathfrak{h}$. The roots span $\mathfrak{h}^*$, and $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] = \mathbb{C}t_\alpha$. Conjugacy makes the Cartan dimension independent of the choice. Lecture 8 builds on this.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 8.1–8.3; the imports are from Sections 5.4, 6.4, 15.3 and 16.4. Erdmann–Wildon, *Introduction to Lie Algebras*, Chapter 10. Exercise Sheet 7 gives further calculations and proofs.